Check for updates

Applied Computational Intelligence and Soft Computing

WILEY

RESEARCH ARTICLE OPEN ACCESS

3D VB Light-UNet With Pooling Indices for MRI Brain Tumor Segmentation on Volumetric Images

Anita Desiani[1] | Deshinta Arrova Dewi[2] | Dite Geovani[3] | Muhammad Arhami[4] | Emy Setyaningsih[5] | Lucky Indra Kesuma[6]

[1]Department of Mathematics, Universitas Sriwijaya, Indralaya, South Sumatera, Indonesia | [2]Department of Computer Science and Technology, INTI International University, Putra Nilai, Negeri Sembilan, Malaysia | [3]Department of Engineering Science, Universitas Sriwijaya, Indralaya, South Sumatera, Indonesia | [4]Department of Informatics Engineering, Politeknik Negeri Lhokseumawe, Lhokseumawe, Aceh, Indonesia | [5]Department of Computer Systems Engineering, Akprind University, Yogyakarta, Special Region of Yogyakarta, Indonesia | [6]Department of Electrical Engineering, Universitas Sriwijaya, Indralaya, South Sumatera, Indonesia

Correspondence: Anita Desiani (anita_desiani@unsri.ac.id)

Received: 2 September 2025 | **Revised:** 29 April 2026 | **Accepted:** 7 May 2026

Academic Editor: Pramita Mishra

Keywords: brain | health risk | innovation | segmentation | tumor

ABSTRACT

Brain tumors are among the most dangerous cancers and have a high death rate. One way to deal with the high rate of brain tumor cases is through early detection of brain tumors on brain MRI images. Early detection of tumors on brain MRI images is carried out by detecting abnormalities in nonenhancing tumor cells, peritumoral edema, and enhancing tumor cells on brain MRI images. This can be achieved through segmentation to obtain the detailed features of these cells on brain MRI images. The CNN architecture that is widely used in three-dimensional image segmentation is the volumetric UNet (V-UNet) architecture. This study proposes the VB Light-UNet architecture to segment nonenhancing tumor, peritumoral edema, and enhancing tumor cell features on brain MRI images. The VB Light-UNet architecture is a V-UNet architecture that removes the bridge part, adds batch normalization at each convolution layer, and uses pooling indices in the max pooling operation. This study still maintains the three-dimensional MRI image without converting it to a two-dimensional image first, to utilize all spatial information in the three-dimensional image. Evaluation of the performance of the VB Light-UNet architecture in brain tumor segmentation on brain MRI images showed accuracy, sensitivity, MCC, and G-mean above 0.92, specificity and Dice score above 0.83, IoU of 0.74, and Cohen's kappa of 0.52. The segmentation results obtained are very similar to the ground truth. These results show that the VB Light-UNet architecture is robust and capable of precisely and accurately segmenting brain tumors on brain MRI images.

1 | Introduction

Brain tumors are among the most dangerous cancers and have a high death rate [1]. Brain tumors are the 10th leading cause of death worldwide [1]. The brain consists of many nerve cells and tissues that control major activities throughout the body, such as breathing, sensing, and moving [2]. Each cell plays a specific role in maintaining normal brain function, including tightly regulated processes such as cell growth and division, which, when disrupted, can lead to tumor formation. Brain tumors are caused by abnormal cell growth in the brain, such as the inability to carry out their functions properly, and the loss of the capacity of these cells [1, 3].

One way to deal with the high rate of brain tumor cases is by early detection of brain tumors on brain MRI images [1]. Brain MRI images are three-dimensional images that show the MRI results of the brain [1]. Early detection of tumors on brain MRI images is

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

Copyright © 2026 Anita Desiani et al. *Applied Computational Intelligence and Soft Computing* published by John Wiley & Sons Ltd.

carried out by detecting abnormalities in nonenhancing tumor cells, peritumoral edema, and enhancing tumor regions on brain MRI images [4]. To obtain detailed features of nonenhancing tumor cells, peritumoral edema, and enhancing tumor cells on brain MRI images, segmentation assistance can be done using segmentation assistance [1]. Image segmentation is a process of identifying important features that will be analyzed from an image [5]. Manual segmentation of three-dimensional MRI brain images is subjective, requires very high precision, and takes quite a long time [6]. MRI images in three dimensions have higher accuracy than two-dimensional images because the spatial information in three-dimensional images is composed of length, width, and depth [7], while the spatial information in two-dimensional images is only length and width [8]. This causes manual segmentation of three-dimensional images to have quite high complexity [6].

The development of automatic segmentation algorithms is needed to facilitate the work of medical experts in image segmentation [9]. The use of convolutional neural network (CNN) methods has been widely developed in the medical world to carry out automatic segmentation with machines [6]. Zhao et al. [6] used a fully connected layer architecture for brain tumor segmentation with a two-dimensional approach. The two-dimensional approach was carried out by converting a two-dimensional image into a two-dimensional one using slice techniques in the axial, coronal, and sagittal planes, obtaining a Dice score of 0.79 and a sensitivity of 0.83. Wang and Chung [4] used the UNet and ResNet architecture for a two-dimensional approach to brain tumor segmentation. The two-dimensional approach was carried out by converting the three-dimensional image into a two-dimensional image using the slice technique in the coronal plane and obtained a sensitivity value of 0.83 and a specificity of 0.9. When a three-dimensional MRI image is converted into a two-dimensional image for segmentation, it offers the advantage of being able to recognize images because of the thin layer, smaller image size, and less memory usage [6]. In image segmentation, converting MRI data from three-dimensional to two-dimensional representations has the potential to reduce segmentation performance. This reduction is caused by the loss of important features and spatial information, particularly depth information, resulting from the dimensional transformation [6]. This transformation leads to a loss of information regarding the correlation between slices [6]. To overcome the resulting limitations in segmentation performance, various studies have proposed architectural approaches that preserve three-dimensional contextual features [10–12].

The use of the CNN for image segmentation commonly relies on 2D convolution; however, applying 2D convolution to three-dimensional image data can lead to the loss of important depth information because it only processes spatial features in the length and width dimensions [13]. As a result, interslice contextual information within the volumetric data may not be fully captured, potentially reducing segmentation accuracy. To overcome this limitation, a CNN architecture employing 3D convolution is required. Unlike 2D convolution, which extracts features only from the length and width dimensions, 3D convolution can simultaneously capture features from length, width, and depth in three-dimensional images [14]. Consequently, 3D convolution-based architectures are more effective in utilizing the full volumetric information of three-dimensional images and can provide accurate segmentation results [14]. The CNN architecture, which is widely used in three-dimensional image segmentation, is the Volumetric UNet (V-UNet) architecture. V-UNet is a UNet architecture packaged using 3D convolution to segment three-dimensional images [15]. The V-UNet architecture was built to utilize features in three directions, namely, length, width, and depth, in three-dimensional images [15]. Sherman [16] used the V-UNet architecture for brain tumor segmentation and obtained a Dice score of 0.89. However, Sherman [16] only segmented the enhancing tumor region. Hua et al. [17] used the V-UNet architecture to segment brain tumors into nonenhancing tumor, peritumoral edema, and enhancing tumor region, but the Dice score obtained was only 0.79.

The V-UNet architecture has a large number of parameters, with more than 35 million parameters [14]. A large number of parameters result in overfitting and longer training time because the weight values are too varied during initialization [14, 18]. Overfitting can be overcome by reducing parameters in the V-UNet architecture [14, 18, 19]. The V-UNet architecture consists of three parts, namely, the encoder, the bridge, and the decoder. The bridge part of the V-UNet architecture uses dense blocks. A dense block is a block that concatenates to combine the results of the previous layer, so it can obtain more information from the input, but results in the number of parameters being larger. Reducing parameters in the V-UNet architecture can be achieved by eliminating the bridge part. However, the bridge also contributes to feature fusion between the encoder and decoder by abstracting deep semantic information. To address this, the proposed VB Light-UNet employs a dual feature fusion mechanism consisting of pooling indices and direct encoder–decoder concatenation. The pooling indices preserve precise spatial details during upsampling, while the direct concatenation allows the decoder to recover informative and contextually relevant features from the encoder, ensuring effective feature fusion even without the bridge structure.

The V-UNet architecture has deep layers, similar to CNN architectures in general. The presence of deep layers and a large number of parameters makes it difficult to initialize the model weights. The generation of highly variable weights can slow the convergence of the training process. One technique that can facilitate weight initialization by preventing excessively wide variation in weight values is batch normalization (BN) [20]. BN is a technique that trains neural networks by normalizing the input to layers for each mini-batch before training to speed up the training process and neural network convergence. BN can make the training process faster and more stable because the range of weights for each learning is maintained as normal [20]. Sun et al. [21] applied BN on the fully convolutional network for brain tumor segmentation and obtained a Dice score and sensitivity of 0.72. However, this study did not measure accuracy.

The V-UNet architecture usually applies a max pooling operation after each convolutional layer in the encoder to reduce the spatial resolution of feature maps by selecting the maximum value within each local patch [15]. While this operation produces a more compact representation, it may also lead to the loss of fine-grained spatial information. In the original V-UNet, max pooling is performed without storing the indices of the selected maximum values. As a result, important spatial details may not be fully recovered during the upsampling process in the decoder, which can affect the model's ability to reconstruct precise feature locations [22]. To address this limitation, pooling indices are

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/1114045 by Bangladesh Hinari NPL, Wiley Online Library on [19/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

utilized to record the locations of the maximum values during the max pooling operation. This approach does not replace max pooling, but augments it by preserving the positional information of selected features. Conceptually, this mechanism is equivalent to the unpooling operation introduced in SegNet, where the stored indices are reused to guide the upsampling process [23]. By leveraging these indices, the decoder can more accurately reconstruct spatial structures and preserve important local information, thereby improving reconstruction quality [24]. The effectiveness of this approach has also been demonstrated in several medical image segmentation tasks, including pelvic region segmentation [24] and breast tumor segmentation [22].

Previous studies have shown that CNN-based architectures achieve promising results in brain tumor segmentation. However, challenges related to computational complexity and efficiency remain. Therefore, this study proposes a VB Light-UNet architecture for the segmentation of three-dimensional MRI brain images. The VB Light-UNet is a modified version of V-UNet that removes the bridge section, integrates BN in each convolutional layer, and employs pooling indices in the max pooling operation. These architectural modifications are designed to enhance segmentation performance while maintaining computational efficiency, particularly for complex 3D brain MRI data. The elimination of the bridge section aims to reduce model parameters and minimize computational overhead without compromising feature extraction capability. The incorporation of BN helps stabilize gradient updates and accelerates model convergence during training, leading to more reliable optimization. In addition, pooling indices are employed to preserve spatial information during downsampling and guide the upsampling process, enabling more accurate recovery of fine structural details. This mechanism is conceptually equivalent to the unpooling operation introduced in SegNet. This study also retains the three-dimensional MRI image format to fully leverage spatial contextual information, including length, width, and depth. Overall, the proposed VB Light-UNet provides a lightweight yet effective framework that improves segmentation efficiency and enhances the reconstruction of brain tumor boundaries compared to the standard V-UNet. The VB Light-UNet architecture performs voxelwise segmentation of 3D brain tumor MRI images into four categories: nonenhancing tumor, peritumoral edema, enhancing tumor, and background. The nonenhancing tumor feature is an area that indicates the presence of a tumor, the peritumoral edema feature is an area that indicates swelling of the tissue around the tumor, the enhancing tumor feature is an area that shows changes in blood circulation during tumor growth, and the background is an area that is not used for brain tumor analysis. The success rate of the proposed VB Light-UNet architecture is measured by calculating architectural performance evaluations such as accuracy, sensitivity, specificity, IoU score, Cohen's kappa, dice score, G-mean, Matthews correlation coefficient (MCC), and ROC. The VB Light-UNet architecture is expected to have good performance evaluation results in three-dimensional image brain tumor segmentation.

2 | Materials and Methods

2.1 | Data Description

The dataset used in this study is Multimodal Brain Tumor Segmentation (BraTS) 2019 and 2020, organized by the Medical Image Computing and Computer-Assisted Intervention (MICCAI) BraTS Challenge [25, 26]. The BraTs dataset contains a total of 2620 3D MRI scans in NifTI (.nii) format. Each MRI volume is accompanied by expert-annotated ground truth masks that delineate tumor subregions, including nonenhancing tumor, peritumoral edema, and enhancing tumor. The dataset naturally contains a larger proportion of background voxels compared to foreground tumor regions, which reflects the real anatomical distribution in MRI brain images. Examples of several original MRI volumes and their corresponding ground truth masks in both 3D and semantic visualization are presented in Table 1. A detailed view of each image feature in the semantic view of the ground truth image of BraTS2020_Training_200.nii can be seen in Figure 1.

2.2 | Image Normalization

Operations on three-dimensional images are performed on each smallest unit, known as a volumetric pixel (voxel). Voxels represent the width, height, and depth of a three-dimensional image. The data preprocessing stage is carried out by normalizing the voxel values so that the voxel values fall within the range of 0–1, using the minimum–maximum scalar method. Normalization is carried out to overcome different data scales and intensity ranges that are too varied. This process impacts the training process by mitigating numerical instability and preventing the model from being biased toward high-magnitude features. By scaling inputs to a uniform range, normalization ensures smoother gradient flow, thereby accelerating the convergence of the optimization algorithm during training. The calculation operation in the minimum–maximum scalar method can be seen in Equation (1).

$$d = b^{l}_{jm} = \frac{x^{l}_{jm} - x^{l}_{jm_{\min}}}{x^{l}_{jm_{\max}} - x^{l}_{jm_{\min}}}, \tag{1}$$

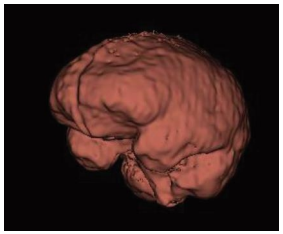
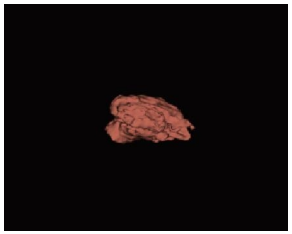
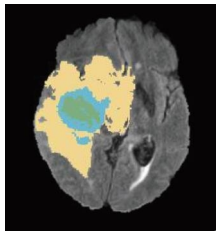
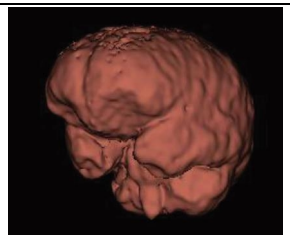
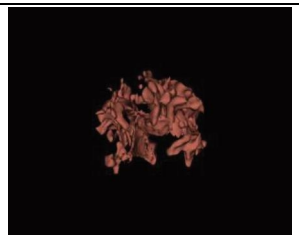
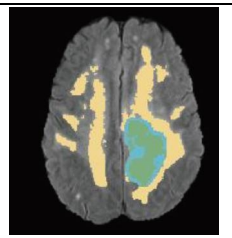
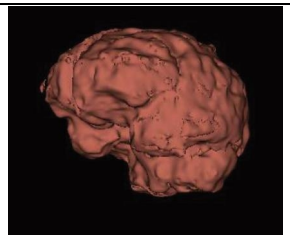
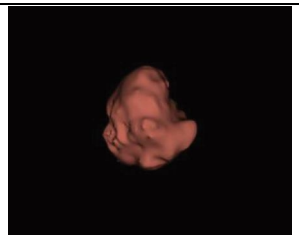
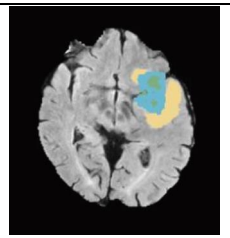
where $\mathbf{l}$ indicates the layer of the image depth, $\mathbf{j}$ indicates the row or image width, $\boldsymbol{m}$ indicates the column or image height, $\mathbf{b^{l}_{jm}}$ is the voxel normalization result at the $\mathbf{l}$-th layer, $\mathbf{j}$-th row, and $\mathbf{m}$-th column, $\mathbf{x^{l}_{jk}}$ is the voxel value at the $\mathbf{l}$-th layer, $\mathbf{j}$-th row, and $\mathbf{m}$-th column to be normalized, and $\mathbf{x^{l}_{jk\,max}}$ and $\mathbf{x^{l}_{jk\,min}}$ are the maximum and minimum voxel values at the $\mathbf{l}$-th layer, $\mathbf{j}$-th row, and $\mathbf{m}$-th column.

2.3 | VB Light-UNet Architecture

The proposed VB Light-UNet architecture is a modified version of the standard V-UNet that removes the bridge section, incorporates BN after each convolutional layer, and utilizes pooling indices in the max pooling operation. These modifications were introduced to address limitations observed in the original V-UNet when handling high-dimensional 3D MRI data, such as high computational cost, unstable gradient flow, and loss of spatial detail during upsampling. The removal of the bridge section reduces the number of trainable parameters and computational complexity while maintaining the network's ability to capture hierarchical features. The integration of BN layers stabilizes the training process, accelerates convergence, and mitigates internal covariate shift during optimization. The use of pooling indices allows the model to retain precise spatial information during max pooling and effectively reconstruct fine-grained details during the upsampling stage. Collectively, these

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/1114045 by Bangladesh Hinari NPL, Wiley Online Library on [19/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 | Images and ground truth of the BraTS dataset.

File name	Image	Ground truth	Semantic ground truth
BraTS2020_Training_001.nii			
BraTS2020_Training_069.nii			
BraTS2020_Training_200.nii			

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/1114045 by Bangladesh Hinari NPL, Wiley Online Library on [19/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

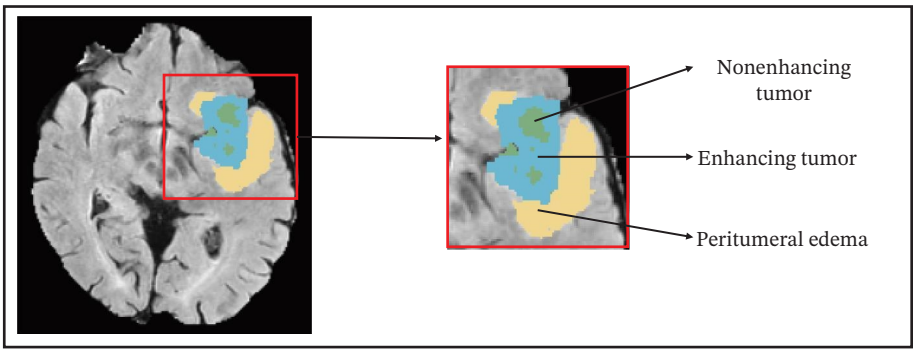


FIGURE 1 | Detailed view of each feature in the semantic ground truth image BraTS2020_Training_200.nii.

architectural adjustments produce a lightweight yet robust network capable of preserving spatial consistency and improving the delineation of brain tumor boundaries in 3D MRI segmentation tasks. The VB Light-UNet architecture can be seen in Figure 2.

Based on Figure 2, the VB Light-UNet architecture consists of two parts, namely, the encoder and decoder. In the encoder, there are four blocks, each consisting of 3D convolution layers with a $3 \times 3 \times 3$ kernel size, BN, Rectified Linear Unit (ReLU) activation function, dropout, and max pooling. The decoder has three blocks, where each block consists of upsampling, 3D convolution layers with a $3 \times 3 \times 3$ kernel size, BN, ReLU activation function, and dropout. The last block in the decoder ends with a 3D convolutional layer with a $1 \times 1 \times 1$ kernel size and a softmax activation function. The results from the encoder are connected directly to the decoder by carrying out a concatenation operation from the end of the encoder to the beginning of the decoder. In the encoder, 3D convolution operation layers with a $3 \times 3 \times 3$ kernel size are carried out using Equation (2).

$$v_{jm}^{l} = b + \left(\sum_{u=0}^{n-1} \sum_{v=0}^{n-1} \sum_{w=0}^{n-1} d_{v+j,w+m}^{u+l} \times k_{v+1,w+1}^{u+1} \right), \tag{2}$$

for $j = 1, 2, \ldots, n$ and $k = 1, 2, \ldots, n$, where n is the number of kernel layers; l indicates the layer of the image depth; j indicates the row or image width; m indicates the column or image height; v_{jk}^{l} is the entry of the voxel convolution matrix at the l-th layer,

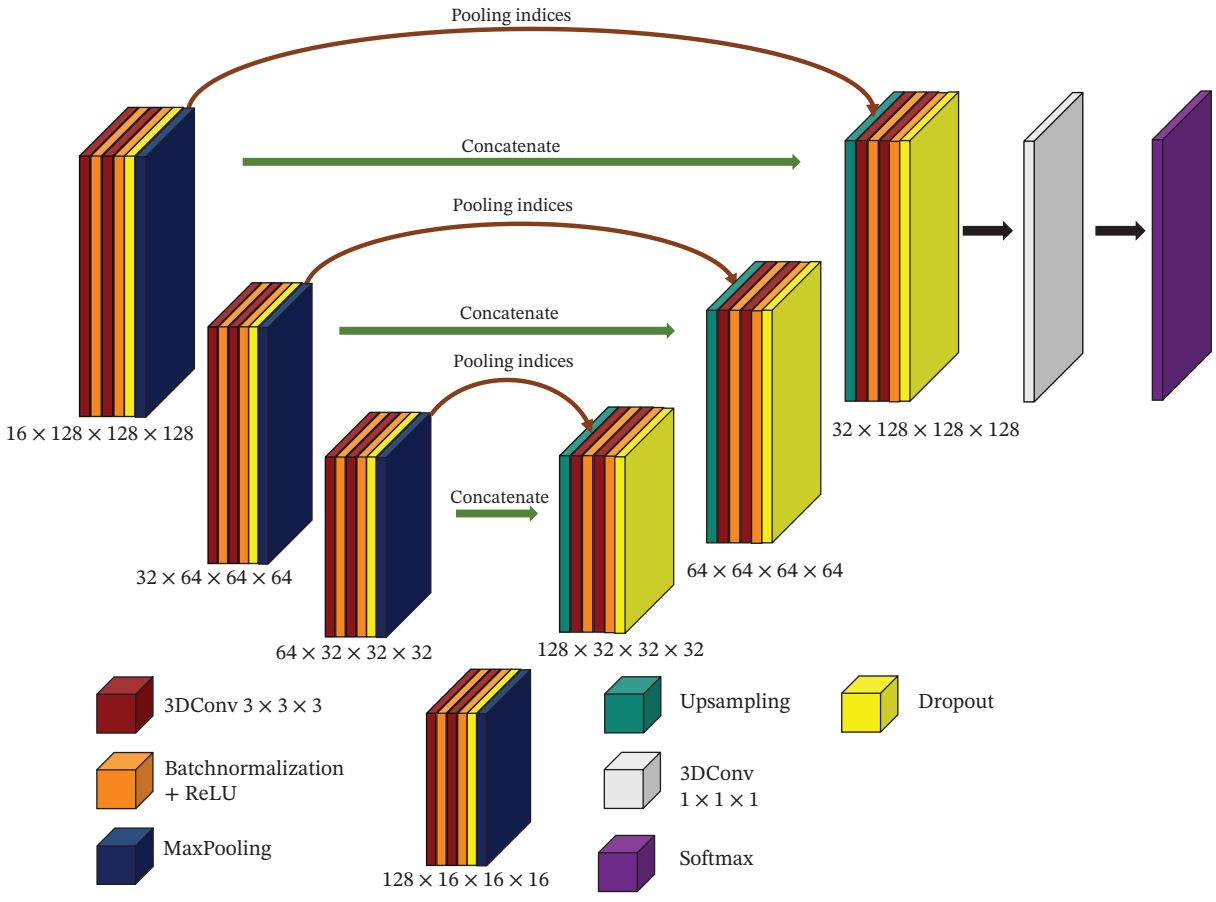


FIGURE 2 | VB Light-UNet architecture.

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/1114045 by Bangladesh Hinari NPL, Wiley Online Library on [19/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

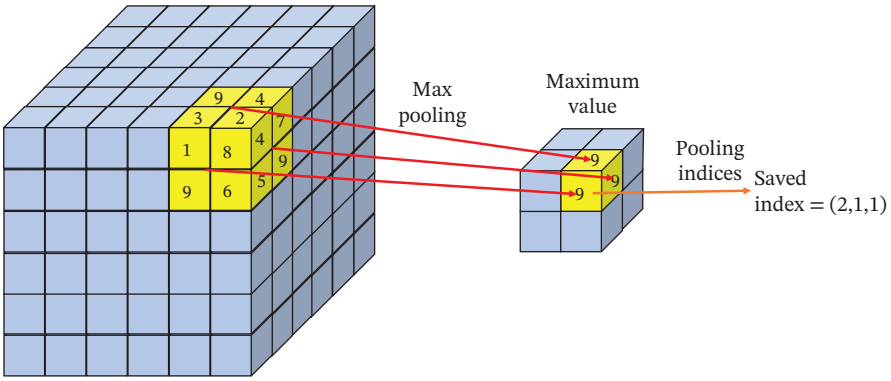


FIGURE 3 | An illustration of the pooling indices process.

j-th row, and m-th column; $d^{u+l}_{v+j,w+k}$ is a voxel matrix entry of the $u+l$-th layer, $v+j$-th row, and $w+m$-th column; $k^{u+1}_{v+1,w+1}$ is the kernel matrix entry of the $u+1$-th layer, $v+1$-th row, and $w+1$-th column; and b is the bias value.

The voxel matrix entry values from the results of the 3D convolution of $1 \times 1 \times 1$ kernel size operation layers with a $3 \times 3 \times 3$ kernel size are normalized using BN, as defined in Equations (3)–(5).

$$\mu_j = \frac{1}{p}\sum_{i=1}^{p} v^l_{jm}, \tag{3}$$

$$\sigma_j^2 = \frac{1}{p}\sum_{i=1}^{p}\left(v^l_{jm} - \mu_j\right)^2, \tag{4}$$

$$\widehat{v}^l_{jm} = \frac{v^l_{jm} - \mu_j}{\sqrt{\sigma_j^2 + \varepsilon}}, \tag{5}$$

where μ_j is the average value of each mini-batch; σ_j^2 is the variance value for each mini-batch; j is the number of mini-batches; p is the amount of data in a mini-batch; $\widehat{v}^l_{jm}$ is the result of the normalization of voxel values in the l-th layer, j-th row, and m-th column; v^l_{jm} is the input voxel matrix entry resulting from the convolution operation on the l-th layer, j-th row, and m-th column; and ε is the smallest constant to avoid division by zero.

The output results from the BN operation are substituted into the ReLU activation function. ReLU is a nonlinear activation function that changes all negative voxel values in the image. The ReLU function is defined in Equation (6).

$$r = R\left(\widehat{v}^l_{jm}\right) = \max\left(0, \widehat{v}^l_{jm}\right), \tag{6}$$

where r is the output result of the ReLU and $\widehat{v}^l_{jm}$ is the input voxel value from the result of the ReLU activation function operation at the l-th layer, j-th row, and m-th column. To improve generalization and reduce overfitting, a dropout layer is subsequently applied to the activated feature map, randomly deactivating a fraction of the neurons during training. The resulting feature map is then forwarded to the max pooling operation for spatial dimension reduction. In the max pooling operation, the index of the maximum value of the max pooling results is saved using pooling indices. The maximum value index from the max pooling results saved in the pooling indices is used in the upsampling process. An illustration of the pooling indices process can be seen in Figure 3.

In Figure 3, the pooling indices process is illustrated. During the max pooling operation in the encoder, a pooling size of $2 \times 2 \times 2$ is applied to the feature maps generated after BN. For each pooling window, the index of the maximum activation is recorded. For example, if the maximum value 9 is located at position (2, 1, 1), then the index is stored. In the VB Light architecture, encoder and decoder features are still merged through concatenation to preserve contextual and semantic information. The pooling indices mechanism is introduced as an additional spatial guidance strategy rather than a replacement for concatenation. During decoding, the upsampling process is guided by the stored indices, allowing activations to be repositioned at their original maximal locations. By combining concatenation-based feature fusion with index-guided spatial reconstruction, the model benefits from both rich contextual representation and precise localization. This complementary mechanism reduces ambiguity caused by interpolation-based upsampling and improves structural consistency, contributing to the lightweight yet effective segmentation performance of the VB Light-UNet architecture. Feature maps that have gone through upsampling are continued with the 3D convolution layers with a $3 \times 3 \times 3$ kernel size, BN, ReLU activation function, and dropout as in the encoder. The results of the operations in the encoder and decoder are combined using concatenate. In the last layer of the decoder, there is a 3D convolution operation with a $1 \times 1 \times 1$ kernel size. At the end of the VB Light-UNet architecture, a softmax activation function calculation operation is performed using Equation (7).

$$s = s(g_m) = \frac{e^{g_m}}{\sum_{n=1}^{N} e^{g_n}}, \quad k = 1, \ \ldots, N, \tag{7}$$

where N is the number of classes, s is the output result of the softmax, and g_m is the input result of BN in the m-th column. The final stage carried out is calculating the loss function using the Dice loss function. The loss function is used to calculate errors in the model by comparing the predicted values and the actual values. The loss function is capable of optimizing the parameters of the model used. The Dice loss function is a loss function that can be used in image segmentation and has been widely used in medical image segmentation. The Dice loss function can be calculated using Equation (8).

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/1114045 by Bangladesh Hinari NPL, Wiley Online Library on [19/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

$$L = 1 - \frac{2\sum_{i=1}^{N} y_i s_i}{\sum_{i=1}^{N} {y_i}^2 + \sum_{i=1}^{N} {s_i}^2 + \varepsilon}, \tag{8}$$

where L is the value of the Dice loss function, s is the predicted output obtained from the softmax activation function, and y is the actual value for each class j, N is the number of rows in the prediction result matrix, and ε is the smallest constant so that the denominator does not become 0. This loss function guides the optimization process by minimizing segmentation errors and improving model accuracy. The Dice loss function inherently mitigates the impact of class imbalance by measuring the overlap between the predicted and ground truth regions rather than the absolute number of voxels per class.

2.4 | Training and Testing

The training process aims to enable the model to learn patterns and recognize data features. The initial stage involves initializing parameters such as the number of epochs and batch size. Before training, the dataset is divided using a percentage split, with 60% for the training set (1700 images) and 40% for the test set (920 images). The training set is split into 70% for training (1190 images) and 30% for validation (510 images). The model weights with the best validation performance are saved and later used for testing data. The validation process is carried out to monitor the model's learning progress and prevent overfitting.

The testing process is the process of implementing the best weights obtained during the training process on the test data. This process measures the success of the proposed VB Light-UNet architecture on new data or data that the model has never recognized. The test results are arranged in the form of a confusion matrix divided into four labels, namely, 0 for the number of background voxels, 1 for the number of tumor cell voxels that do not experience contrast enhancement, 2 for the number of peritumoral edema cell voxels, and 3 for the number of tumor cell voxels that experience contrast enhancement. The results of brain tumor image segmentation will be compared with the ground truth to measure the performance of the proposed VB Light-UNet architecture.

2.5 | Performance Evaluation

Performance evaluation is represented in the form of a matrix called the confusion matrix, which is produced at the testing stage. In the confusion matrix, the actual classes are represented as matrix columns, and the predicted classes are represented as matrix rows. Some of the terms contained in the confusion matrix include true positive (TP), true negative (TN), false positive (FP), and false negative (FN). This study evaluates architectural performance using the confusion matrix by calculating accuracy, sensitivity, specificity, IoU score, Cohen's kappa, Dice score, G-mean, and MCC. These metrics were computed using Equations (9)–(16).

$$\text{Acc} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FN} + \text{FP} + \text{TN}}, \tag{9}$$

$$\text{Sen} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \tag{10}$$

$$\text{Spe} = \frac{\text{TN}}{\text{TN} + \text{FP}}, \tag{11}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}, \tag{12}$$

$$K = \frac{P_o - P_c}{1 - P_c}, \tag{13}$$

$$\text{Dc} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}, \tag{14}$$

$$G - \text{Mean} = \sqrt{\text{Sen} \times \text{Spe}}, \tag{15}$$

$$\text{MCC} = \frac{(\text{TP} \times \text{TN}) - (\text{FP} \times \text{FN})}{(\text{TP} \times \text{FP})(\text{TP} \times \text{FN})(\text{TN} \times \text{FP})(\text{TN} \times \text{FN})}, \tag{16}$$

where P_o is the percentage of the number of voxels in the segmentation results, P_C is the number of possible levels of change in the number of voxels, TP is the proportion of positive cases predicted as positive cases, TN is the proportion of negative cases predicted as negative cases, FP is the proportion of negative cases predicted as positive cases, and FN is the proportion of positive cases predicted as negative cases. An illustration of all the stages in this study can be seen in Figure 4.

3 | Results and Discussion

3.1 | Normalization

Normalization is carried out to overcome different data scales and intensity ranges that vary too much. The normalization method used is a minimum–maximum scalar. This stage aims to ensure that the image pixels are in the same range, namely, between 0 and 1, using Equation (1). The normalization stage also has an important effect on model performance, as it stabilizes the training process and accelerates convergence by preventing large variations in voxel intensity values. By ensuring that all MRI scans have a consistent intensity distribution, the model can focus on learning relevant spatial and structural features rather than differences in brightness or contrast between images.

3.2 | Implementation of VB Light-UNet Architecture

At this stage, the VB Light-UNet architecture will be implemented. The VB Light-UNet architecture is a V-UNet architecture that removes the bridge part, accompanied by the addition of BN at each convolution layer. The aim of eliminating the bridge in the VB Light-UNet architecture is to make the architecture lighter. BN is applied to facilitate weight initialization during the training process, enabling faster convergence and reducing training time. Figure 2 shows the VB Light-UNet architecture, which consists of two parts, namely, the encoder and decoder.

In the encoder section of the VB Light-UNet architecture, four blocks are employed, each consisting of 3D convolution layers with a $3 \times 3 \times 3$ kernel size, BN, ReLU activation function, dropout, and max pooling. The input image with a size of $128 \times 128 \times 128$ voxels is processed in the first encoder block. In this block, the first 3D convolution layer applies 64 filters with a kernel $3 \times 3 \times 3$ size as illustrated in the red box. The output feature maps are subsequently normalized using BN followed by a ReLU activation function, as shown in the orange box. The

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/1114045 by Bangladesh Hinari NPL, Wiley Online Library on [19/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

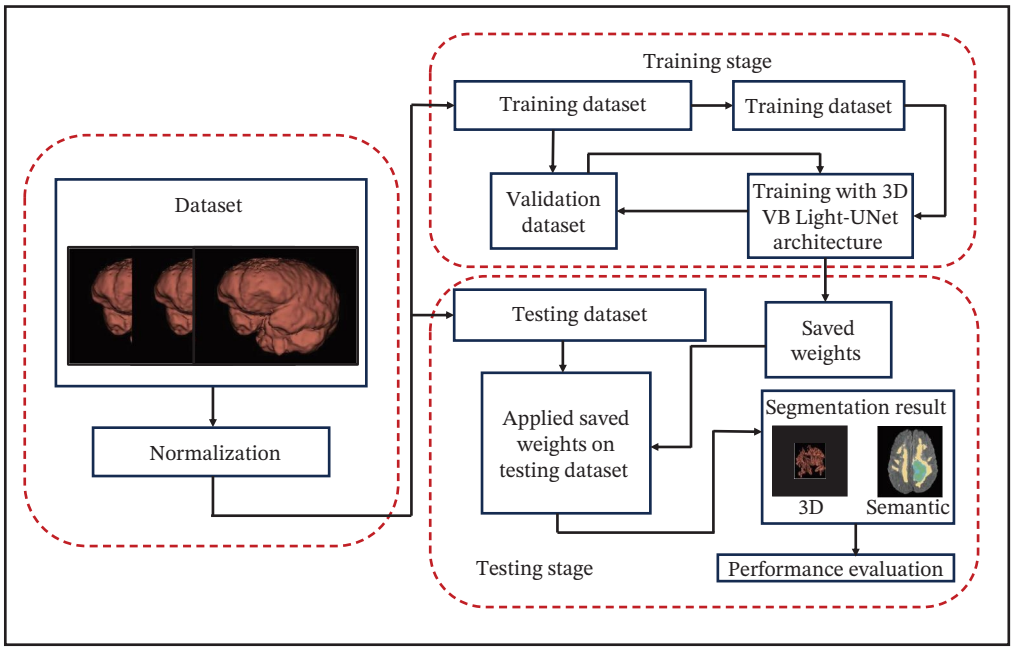


FIGURE 4 | The proposed method for brain tumor segmentation using VB Light-UNet architecture.

second convolution layer in the same block performs an identical operation, consisting of a 3D convolution with a $3 \times 3 \times 3$ kernel size, BN, and ReLU activation function. The output of the last convolution layer in the first block is then passed through a dropout operation, as illustrated by the yellow box. Subsequently, the resulting feature map is downsampled using a max pooling operation, as shown by the dark blue box. During the max pooling process, pooling indices are stored to record the spatial locations of the maximum values. The same architectural pattern is applied to the second, third, and fourth encoder blocks, with the number of filters increased to 128, 256, and 512, respectively.

The decoder section of the VB Light-UNet architecture consists of three blocks, each comprising an upsampling operation, followed by 3D convolution layers with a $3 \times 3 \times 3$ kernel size, BN, ReLU activation function, and dropout. In the first decoder block, the feature maps are upsampled to increase their spatial resolution, as shown in the green box. The upsampled feature maps are then processed by a 3D convolution layer with 128 filters and a kernel $3 \times 3 \times 3$ size, followed by a BN and ReLU activation function, as indicated in the red and orange boxes. A second convolution layer with the same configuration is subsequently applied. This is followed by a dropout operation, illustrated by the yellow box. The same decoding process is repeated in the second and third decoder blocks using 64 and 32 filters, respectively. The final decoder block ends with a 3D convolution layer using a $1 \times 1 \times 1$ kernel size, as shown in the white box, followed by a softmax activation function in the purple box to generate the final segmentation output. Skip connections are established between the encoder and decoder sections by concatenating the corresponding feature maps from the encoder to the decoder at each resolution level.

3.3 | Training

The training process is carried out using the VB Light-UNet architecture and produces a VB Light-UNet model. This model has been trained with 1700 training data, which is divided into two parts, namely, 1190 as training data and 510 as validation data. The parameters used in the training process using the VB Light-UNet architecture include 100 epochs, 32 of batch size, and 0.00001 of learning rate for Adam optimization. The selection of these hyperparameters was determined empirically through several preliminary experiments to achieve stable convergence and optimal validation performance. A batch size of 32 provides a balance between computational efficiency and model generalization, while the small learning rate allows gradual weight updates that prevent overshooting during optimization. The training process is carried out by initializing the weights for the first epoch and then calculating the loss for each epoch. The weight for each epoch will be updated if the resulting loss is greater than the loss in the previous epoch. The graph of loss and accuracy for the training and validation data obtained at the training stage can be seen in Figure 5.

In Figure 5(a), it can be seen that the graph of loss during the training process always decreases with each epoch. The loss in validation data experiences increases and decreases. In the first epoch, the loss was above 0.8 and then decreased to below 0.2. In Figure 5(b), it can be seen that the accuracy graph of the training process has increased at each epoch. The training accuracy experienced a slight decrease at the 30th epoch, followed by an increase and another decline at the 50th epoch, before eventually increasing again and stabilizing in the subsequent epochs. In the first epoch, the accuracy was below 0.5, and it gradually increased in the following epochs, exceeding 0.9 and approaching 1. In addition to accuracy, the performance of the model based on the VB Light-UNet architecture was also evaluated using sensitivity and specificity metrics. Sensitivity measures the ability of the VB Light-UNet architecture to identify pixels that belong to the feature to be segmented (foreground). Specificity shows the ability of the VB Light-UNet architecture to identify pixels that belong to background features. A graph of the sensitivity and specificity for the training and validation data obtained at the training stage can be seen in Figure 6.

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/1114045 by Bangladesh Hinari NPL, Wiley Online Library on [19/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

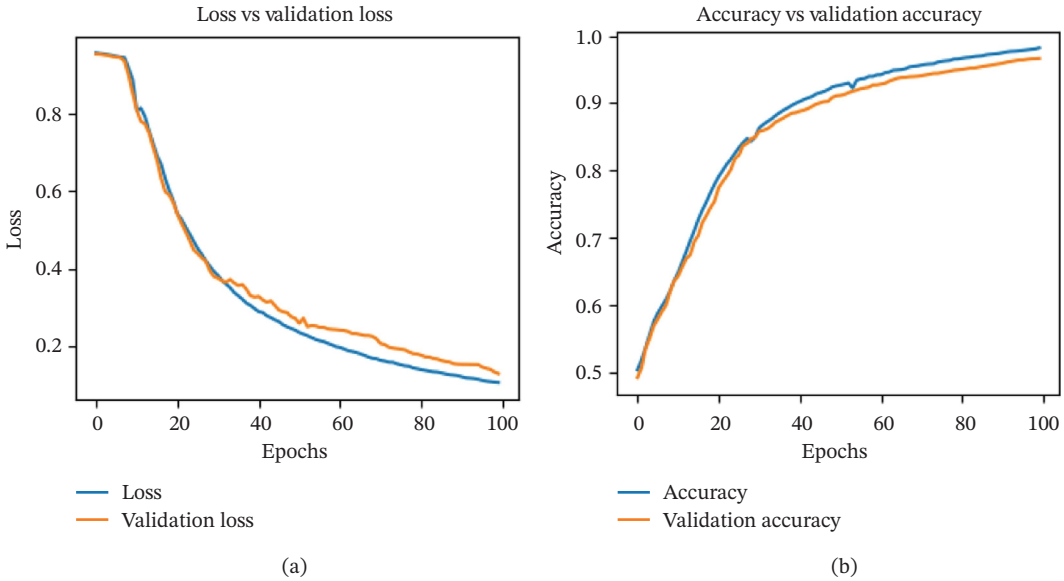


FIGURE 5 | Graph of (a) loss and (b) accuracy in the VB Light-UNet architecture training process.

In Figure 6(a), the sensitivity of the training data shows a steady increase, while the validation data exhibit a slight decrease at the 30th epoch, followed by a stable increase until the final epoch. The sensitivity in the first epoch is below 0.2 and subsequently rises to above 0.8. In Figure 6(b), the specificity of both the training and validation data shows a steady increase, starting below 0.3 in the first epoch and reaching approximately 0.8. Other performance measures recorded during the training process using the VB Light-UNet architecture include the Dice score and IoU score. The Dice score measures the overlap between segmentation result pixels and ground truth pixels, while the IoU score evaluates the ratio between the intersection and the union of the segmented region and the ground truth. The trends of the Dice and IoU scores for both training and validation data are presented in Figure 7.

In Figure 7(a), the Dice score for both training and validation data shows a steady increase throughout the training process, rising from below 0.2 in the first epoch to approximately 0.8 by the final epoch. In Figure 7(b), the IoU score also demonstrates a consistent increase, starting below 0.1 and reaching approximately 0.7 at the final epoch. The error rate during the training process is measured using RMSE, which quantifies the difference between the segmentation results and the ground truth. The RMSE trends for training and validation data are shown in Figure 8. The error decreases steadily over the epochs, with values above 0.4 in the first epoch declining to below 0.15 by the final epoch. An RMSE value that is close to 0 indicates that the error level between the segmentation image and the ground truth is very small, so the level of similarity between the segmentation image and the ground truth is higher. Although semantic

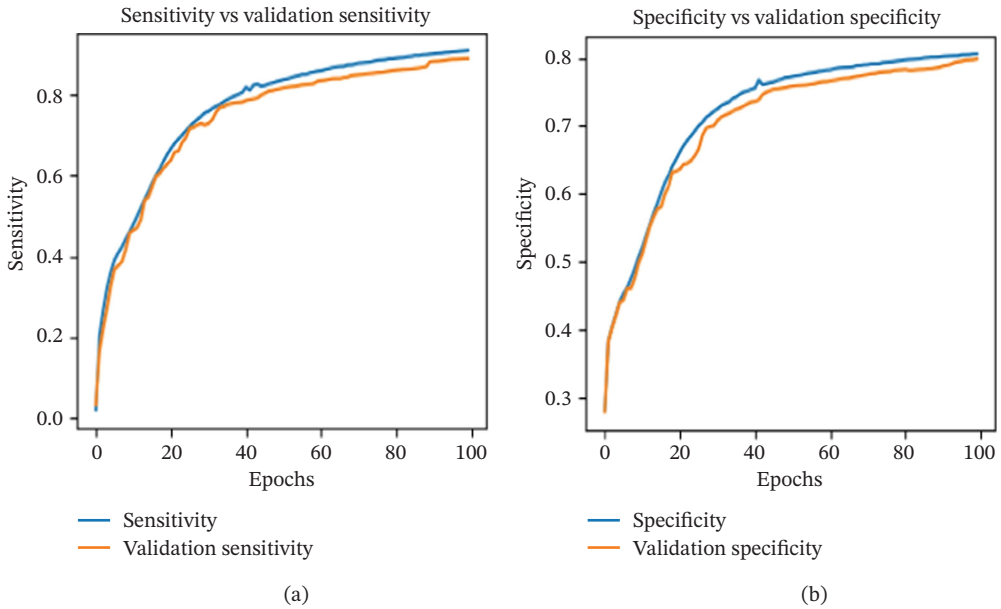


FIGURE 6 | Graph of (a) sensitivity and (b) specificity in the VB Light-UNet architecture training process.

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/1114045 by Bangladesh Hinari NPL, Wiley Online Library on [19/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

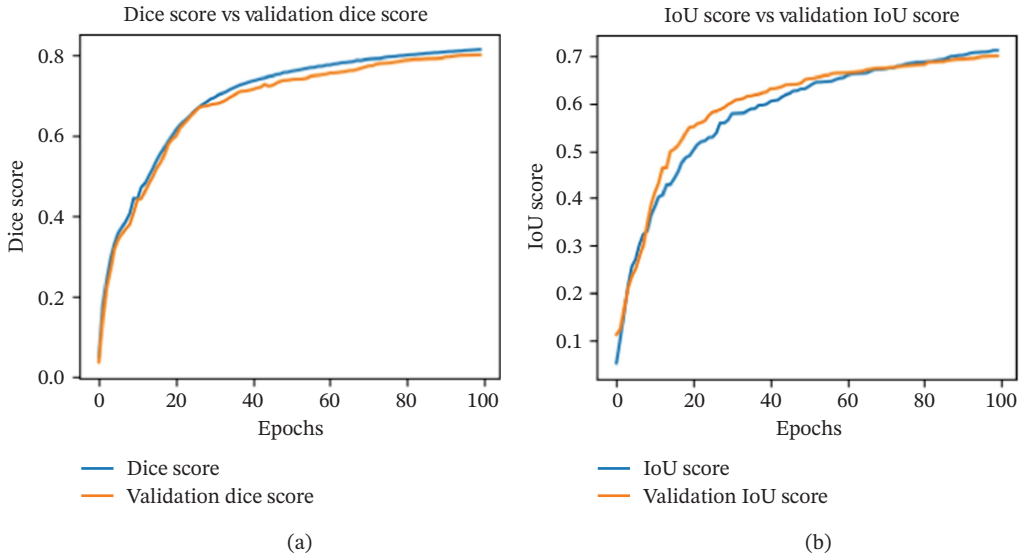


FIGURE 7 | Graph of (a) Dice and (b) IoU score in the VB Light-UNet architecture training process.

segmentation is a pixelwise classification task, RMSE was used in this study as a supplementary metric to observe how far the predicted pixel values deviate from the ground truth, especially since the model outputs soft probabilities before thresholding. However, IoU and Dice scores were also measured to more accurately reflect classification performance. In the training process, the best weights are saved to be used in the testing stage.

3.4 | Testing

At the testing stage, predictions are made based on the best weight results obtained at the VB Light-UNet architecture training stage. The testing data obtained from the split data are 40% of the total images or 920 images. At the testing stage, an image of the brain tumor segmentation prediction results is obtained so that the model performance can be evaluated. The brain tumor segmentation prediction results were compared with the ground truth. Several comparisons of the original image, ground truth, and segmentation results can be seen in Table 2.

Table 2 shows the file name, original image view, three-dimensional view of ground truth and segmentation results, as well as the semantic view of ground truth and segmentation results. The brain tumor segmentation results obtained are similar to the ground truth. In the three-dimensional view column, the brain tumor segmentation results show a brain tumor cell without clearly viewing the nonenhancing tumor, peritumoral edema, and enhancing tumor features contained in the brain tumor cells. The detailed view of nonenhancing tumor, peritumoral edema, and enhancing tumor features in segmented brain tumor cells can be seen in the semantic view column of the ground truth and segmentation results. In the semantic view column, a clear view of each feature contained in the tumor cells is given in the three-dimensional view column. Semantic view columns provide a detailed and clear view of each feature of brain tumor cells. In the image in the semantic view column, the yellow area is labeled peritumoral edema, the blue area is the enhancing tumor label, the green area is the nonenhancing tumor label, and the other areas are background. Based on the semantic view column, it can be seen that the segmentation results obtained were successful in separating nonenhancing tumor, peritumoral edema, and enhancing tumor features in brain tumor cells. However, unfortunately, there is a slight discrepancy in the predicted area, such as in the BraT-S2020_Training_365.nii image, which predicts the area slightly exceeds the size that it should be found in the ground truth. The discrepancies in predicted segmentation areas may be caused by variations in tumor boundary contrast, noise levels, and the model's tendency to overfit certain regions with similar intensity patterns. Future improvements can include incorporating attention mechanisms or boundary refinement modules to better capture local edge information and reduce oversegmentation. Based on the segmentation results obtained, the performance evaluation of the VB Light-UNet model will be measured using a confusion matrix. In the resulting confusion matrix, there are four labels, namely, nonenhancing tumor, peritumoral edema, enhancing tumor, and background. Evaluation of the performance of the VB Light-UNet model, which is calculated based on the confusion matrix obtained, including the accuracy, sensitivity, specificity, IoU score, Cohen's Kappa, Dice score, G-mean, and MCC.

In this study, the VB Light-UNet architecture achieved a high pixel accuracy of 0.98. However, given the inherent class imbalance between the small tumor regions (foreground) and the extensive nontumor tissue (background), accuracy alone can be misleading. Therefore, we prioritized metrics that evaluate the balance between classes. The model demonstrated robust performance with a G-mean of 0.93, which indicates a geometric balance between the classification of the majority (background) and minority (tumor) classes. This is further supported by an MCC of 0.96, confirming a strong correlation between the prediction and ground truth even in the presence of imbalanced data. Specifically, regarding class detection, the model achieved a Sensitivity of 0.96, ensuring that the majority of actual tumor pixels were correctly identified, while a specificity of 0.89 indicates the model's ability to correctly distinguish nontumor tissue. The segmentation quality is validated by a Dice score of 0.84 and an IoU of 0.74, reflecting substantial overlap between the predicted segmentation and the ground truth. Collectively,

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/1114045 by Bangladesh Hinari NPL, Wiley Online Library on [19/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

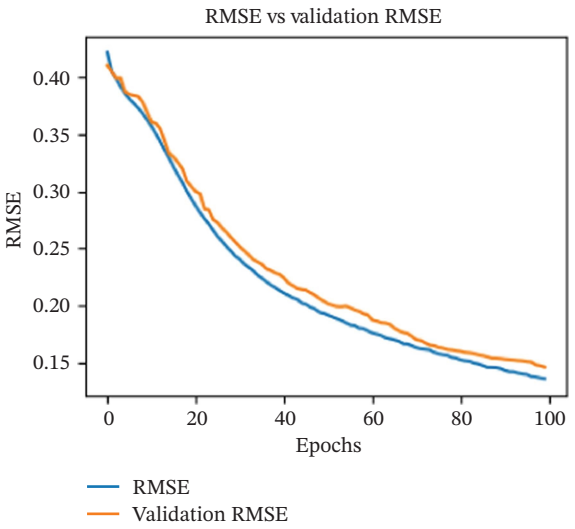


FIGURE 8 | Graph of RMSE in the VB Light-UNet architecture training process.

these metrics confirm that the VB Light-UNet is not biased toward the background but effectively segments tumor regions with high precision. Cohen's Kappa for this study is 0.52, which falls within the moderate agreement category commonly referenced in machine learning and medical imaging evaluation [27, 28]. This result indicates that the model achieves agreement beyond chance in multiclass 3D tumor segmentation. When considered alongside complementary metrics such as Dice score, IoU, G-mean, and MCC, Cohen's kappa provides a more comprehensive assessment of the model's performance and consistency across tumor subregions.

Another performance evaluation measure used in this study is the receiver operating characteristic (ROC) curve. The ROC curve is a two-dimensional plot that illustrates the relationship between the true-positive rate (TPR) and the false-positive rate (FPR). The overall performance of the model is quantified by calculating the area under the curve (AUC) of the ROC curve. A higher AUC value indicates better classification performance of the model. Figure 9 presents the ROC curve and AUC results obtained from applying the training weights of the VB Light-UNet architecture to the test data. Figure 9 shows the AUC results for label 1 (nonenhancing tumor) of 0.94, for label 2 (peritumoral edema) of 0.97, and for label 3 (enhancing tumor) of 0.96. These AUC results indicate that the VB Light-UNet model has excellent performance in segmenting brain tumors for each label because each AUC result is close to 1. The ROC curves for all labels are consistently located near the upper-left corner of the ROC space, indicating a high TPR while maintaining a low FPR. In particular, the highest AUC value obtained for peritumoral edema cells suggests that the model is highly effective in distinguishing edema regions from surrounding tissues, which is clinically important due to the ambiguous boundaries commonly observed in such regions. Overall, the high AUC values close to 1 confirm that the VB Light-UNet model provides reliable and robust segmentation performance for multiclass brain tumor segmentation.

3.5 | Discussion and Analysis

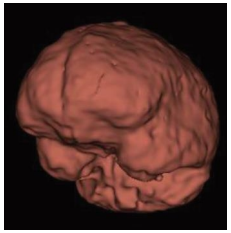
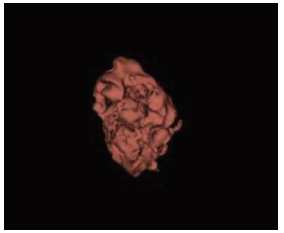
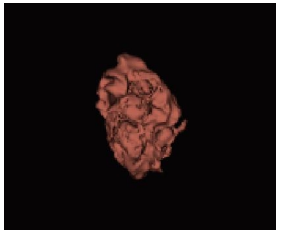
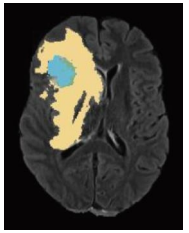
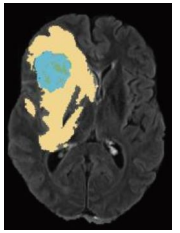
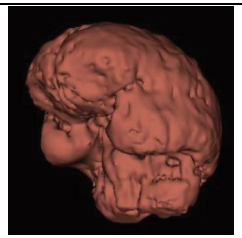
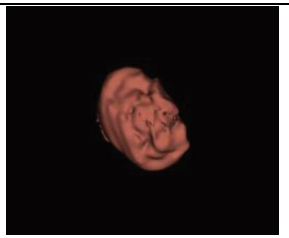
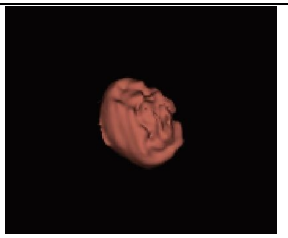
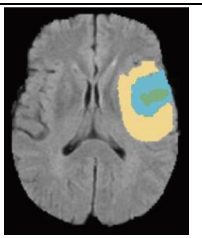
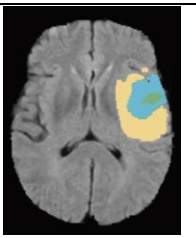
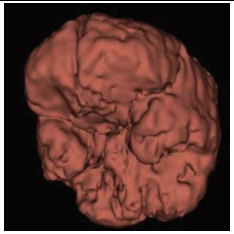
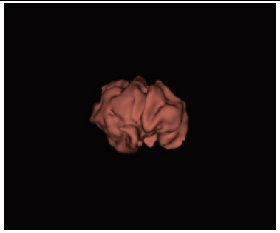
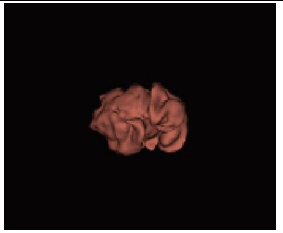
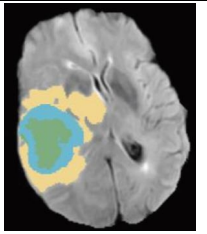
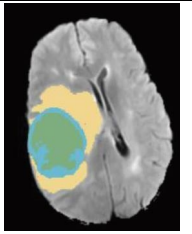
This section provides a deeper interpretation of the results presented in the previous subsections by comparing the proposed VB Light-UNet with the baseline architectures and related studies. While the testing section reported the quantitative performance of each model, this discussion focuses on analyzing the differences in model behavior, architectural efficiency, statistical significance, and the implications of these findings for multiclass 3D brain tumor segmentation. A comparison between the proposed VB Light-UNet and the baseline architectures can be seen in Table 3. The comparison is made on the number of parameters, training time, and inference time. In addition, segmentation performance evaluations, accuracy (Acc), sensitivity (Sen), specificity (Spe), Dice score (Dc), G-mean, IoU, Cohen's kappa (K), and MCC, are also compared.

Based on Table 3, the proposed VB Light-UNet demonstrates a clear and consistent improvement in both computational efficiency and segmentation performance compared to the baseline V-UNet and its intermediate variants. In terms of model complexity, VB Light-UNet reduces the number of parameters from 59.78 million to 22.45 million, corresponding to an approximate 62% reduction. This substantial decrease confirms that the removal of the bridge component, combined with targeted architectural refinements, effectively eliminates redundant computations without compromising the model's representational capacity. Despite this reduction, VB Light-UNet maintains competitive training efficiency, requiring 18,896 s, which is less than half of the original V-UNet training time. Although slightly higher than the bridge-removed variant, this additional cost is justified by the integration of BN, dropout, and pooling indices, which enhance feature stability and generalization. This indicates that the proposed design is not merely simplified but strategically optimized to improve learning robustness. From a performance perspective, VB Light-UNet consistently achieves the best results across all key evaluation metrics. The model attains the highest accuracy, sensitivity, and specificity, demonstrating strong capability in identifying both positive and negative classes. The highest Dice and IoU indicate superior spatial overlap and more precise boundary delineation, which are critical in medical image segmentation involving complex tumor structures. Furthermore, the G-mean value of 0.93 highlights the model's ability to maintain a balanced performance between sensitivity and specificity, suggesting robustness under class imbalance conditions. The Cohen's kappa score of 0.52 indicates a moderate level of agreement beyond chance, which remains acceptable for multiclass brain tumor segmentation tasks where class boundaries are inherently ambiguous.

In terms of computational efficiency during inference, VB Light-UNet achieves an inference time of 120 ms per volume for a 3D input of size $128 \times 128 \times 128$, significantly outperforming the original V-UNet and substantially improving upon the bridge-removed variant. This corresponds to approximately a 3.6× speedup, highlighting the practical significance of the proposed model in real-world scenarios. Although marginally slower than the pooling-indices-only variant, the proposed model delivers considerably better segmentation performance, indicating a more optimal trade-off between speed and accuracy. Overall, the results presented in Table 3 confirm that VB Light-UNet is not merely a simplified architecture but a carefully optimized model that effectively balances performance and efficiency. By reducing architectural complexity and incorporating stabilizing mechanisms, the proposed method achieves superior segmentation results with significantly reduced computational cost,

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/1114045 by Bangladesh Hinari NPL, Wiley Online Library on [19/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 | Comparison of original images, ground truth, and brain tumor segmentation results based on MRI images.

File name	Image	3D view		Semantic view	
		Ground truth	Segmentation result	Ground truth	Segmentation result
BraTS2020_Training_026.nii					
BraTS2020_Training_251.nii					
BraTS2020_Training_365.nii					

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/1114045 by Bangladesh Hinari NPL, Wiley Online Library on [19/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

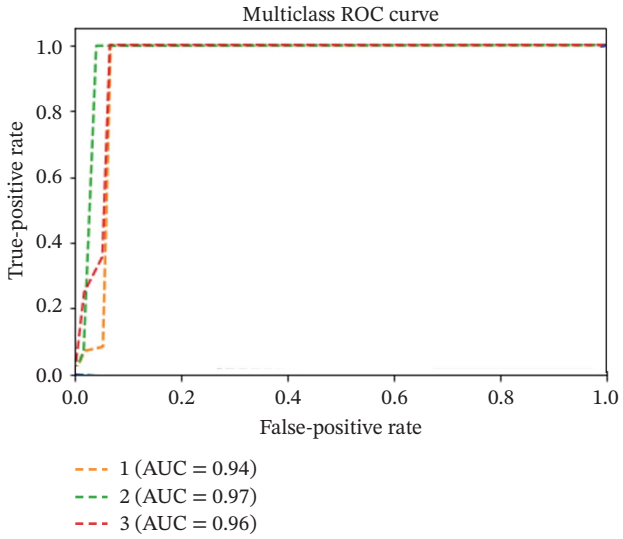


FIGURE 9 | ROC curve of brain tumor segmentation with VB Light-UNet architecture.

making it a strong and practical candidate for real-world three-dimensional brain tumor segmentation applications.

To further address the contribution of each architectural component, a more detailed ablation study is conducted, as presented in Table 3. Specifically, the effects of BN and pooling indices are evaluated independently on top of the V-UNet without the bridge configuration. The inclusion of BN leads to a noticeable improvement in overall performance. Compared to the bridge-removed baseline, the model with BN achieves higher accuracy and sensitivity, indicating improved feature normalization and more stable training dynamics. Additionally, the inference time is reduced, suggesting that BN contributes not only to performance stability but also to computational efficiency. On the other hand, the use of pooling indices demonstrates a different behavior. While this modification achieves the fastest inference time and slightly improves sensitivity, it results in a noticeable decline in segmentation quality, particularly in dice score and MCC. A direct comparison between standard transposed convolution and pooling indices further highlights this trade-off. Although pooling indices improve computational efficiency, the use of pooling indices can significantly degrade segmentation performance, suggesting that learned upsampling is more effective in reconstructing fine spatial details in complex 3D medical images. When both components are integrated in the proposed VB Light-UNet, the model achieves the best overall performance across all evaluation metrics. Significant improvements are observed in Dice score, IoU, G-mean, and Cohen's kappa, demonstrating that the combination of BN and pooling indices provides complementary benefits. BN stabilizes feature distribution, while pooling indices enhance spatial detail recovery, resulting in a more discriminative and robust model. These findings confirm that the proposed architectural design is not merely a combination of modifications, but a synergistic integration where each component contributes differently to the final performance. Therefore, the ablation study clearly justifies the design choices in VB Light-UNet and highlights the importance of combining normalization and spatial preservation mechanisms in three-dimensional medical image segmentation.

To further validate the reliability of the observed performance differences among the evaluated architectures, statistical significance testing was conducted. This analysis was performed to ensure that the improvements achieved by the proposed VB Light-UNet were not due to random variations in the dataset, but reflected consistent and systematic performance gains. The statistical evaluation included the Kolmogorov–Smirnov test to assess the normality of the metric distributions, followed by the Friedman test to determine overall differences across models, and the Wilcoxon signed-rank test for pairwise comparisons.

The Kolmogorov–Smirnov test was conducted to evaluate whether the distribution of each performance metric followed a normal distribution. In this analysis, the null hypothesis (H_0) states that the data are normally distributed, while the alternative hypothesis (H_1) indicates deviation from normality. The results of this test determine the appropriate statistical approach for subsequent hypothesis testing, ensuring that the performance comparison among the evaluated architectures is based on valid assumptions. A summary of the Kolmogorov–Smirnov test outcomes can be seen in Table 4.

In Table 4, the Kolmogorov–Smirnov test results show that the performance metric distributions across the three architectures do not fully satisfy the assumption of normality. For the V-UNet architecture, only the val_sen metric has a p value greater than 0.05, indicating a normal distribution. In contrast, val_spe and val_dice exhibit p values below 0.05, suggesting that these metrics are not normally distributed. A similar pattern appears in the V-UNet without bridge architecture. The val_spe metric meets the normality assumption, while val_sen and val_dice do not, as indicated by their low p values. This implies that the data distribution for this architecture is predominantly non-normal. For the VB Light-UNet model, the validation metrics val_sen, val_spe, and val_dice yielded p values well below the 0.05 significance threshold.

This clearly indicates that none of the metrics follow a normal distribution. Since the Kolmogorov–Smirnov test indicated that most performance metrics did not follow a normal distribution, nonparametric statistical tests were selected for further analysis. The Friedman test was first applied to evaluate whether there were significant performance differences among the three architectures across all evaluation metrics. When the Friedman test indicated significant differences, pairwise Wilcoxon signed-rank tests were subsequently performed to determine which specific model pairs exhibited statistically significant improvements. A summary of the Friedman and Wilcoxon test results is presented in Table 5.

Based on Table 5, the Friedman test was first applied to evaluate whether there were statistically significant differences in performance across the three architectures: V-UNet, V-UNet without bridge, and the proposed VB Light-UNet. The Friedman test yielded p values below 0.001 for sensitivity, specificity, and Dice score, indicating that the three models do not belong to the same performance distribution and that the observed differences are statistically meaningful. Furthermore, to identify where these differences occur, pairwise comparisons using the Wilcoxon signed-rank test were performed. The results showed that all pairwise comparisons were statistically significant ($p < 0.001$ in almost all cases), including all pairwise comparisons. This confirms that each architectural modification,

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/1114405 by Bangladesh Hinari NPL, Wiley Online Library on [19/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3 | Comparison between the proposed VB Light-UNet and the baseline architectures.

Method	Parameters (M)	Training time (s)	Performance evaluation							Inference time (ms)
			Acc	Sen	Spe	Dc	G-mean	IoU	*K*	MCC	
V-UNet	59.78	43,198	0.96	0.93	0.88	0.8	0.9	0.68	0.48	0.94	435
V-UNet without bridge	22.39	14,468	0.94	0.89	0.87	0.76	0.88	0.65	0.3	0.89	265
V-UNet without bridge + batch normalization	22.45	18,896	0.97	0.94	0.88	0.39	0.91	0.63	0.27	0.92	132.1
V-UNet without bridge + pooling indices	22.39	14,468	0.95	0.95	0.84	0.29	0.89	0.67	0.12	0.93	126
VB Light-UNet (proposed method)	22.45	18,896	**0.98**	**0.96**	**0.89**	**0.84**	**0.93**	**0.74**	**0.52**	**0.96**	**120**

Note: The bold values indicate the best performance obtained among all compared methods in each evaluation metric.

removing the bridge and introducing the enhancements in VB Light-UNet, produces a measurable and statistically reliable difference in segmentation performance. The Wilcoxon test results consistently favor VB Light-UNet, which achieved the highest sensitivity, specificity, and Dice score among the three models. Combined with its strong performance in related metrics such as IoU, G-mean, and MCC, the statistical evidence reinforces that the improvements introduced in VB Light-UNet are not only numerically superior but also statistically significant. These findings indicate that the proposed architecture provides genuinely enhanced segmentation capability rather than accidental or noise-driven improvements. Overall, the integrated statistical analysis demonstrates that VB Light-UNet is the highest-performing model with a statistically validated advantage over its baseline counterparts. This confirms that the architectural refinements, removal of the bridge, incorporation of BN and dropout, and the use of pooling indices lead to consistently better segmentation performance across all evaluated metrics.

The success rate of the performance of the VB Light-UNet architecture in segmenting brain tumors in three-dimensional images was compared with the performance results of other architectures. Comparison of the average values of accuracy, sensitivity, and specificity of each feature in this study with other studies can be seen in Table 6. Based on Table 6, brain tumor segmentation using the proposed VB Light-UNet architecture obtained the highest average values of accuracy and sensitivity. The highest specificity value was obtained by Sun et al. [21] and Henry et al. [31], which is 0.9. However, these studies did not measure accuracy, and the sensitivity obtained by both studies remains below 0.85. In addition, the study by Sun et al. [21] performed general brain tumor feature segmentation, specifically enhancing tumor, tumor core, whole tumor, and background. Even though it does not provide the highest specificity value, this study obtained a specificity value close to 0.9, which shows that the architecture works well in predicting background voxels.

The accuracy obtained shows that the accuracy of the predicted voxels is excellent [34]. The resulting sensitivity and specificity values show that the model is excellent at predicting foreground and background voxels. Other performance evaluations compared with other studies include Dice score, G-mean, and IoU score. Comparison of the average values of Dice score, G-mean, and IoU of each feature in this study with other studies can be seen in Table 7. Based on Table 7, brain tumor segmentation using the proposed VB Light-UNet architecture obtained the highest average values of Dice score, G-mean, and IoU score. The Dice score and G-mean values obtained show that the voxels resulting from segmentation have very good agreement with the ground truth voxels, and the model has a very good balance between foreground and background voxels. The IoU scores obtained show that the model can evaluate overlapping areas between segmentation results and ground truth [34]. Other performance evaluations compared with other studies include Cohen's kappa and MCC. The comparison of the average Cohen's kappa (K) and MCC values obtained from each feature in this study with other studies can be seen in Table 8.

In Table 8, it can be seen that brain tumor segmentation using the proposed VB Light-UNet architecture obtained the highest average MCC. The highest Cohen's Kappa value was obtained by

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/1114045 by Bangladesh Hinari NPL, Wiley Online Library on [19/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4 | Kolmogorov–Smirnov test results.

Method	Metric	KS-stat	*p* value	Distribution
V-UNet	Val_sen	0.0454	0.2359	Normal
	Val_spe	0.0632	0.0324	Not normal
	Val_dice	0.0704	0.0121	Not normal
V-UNet without bridge	Val_sen	0.1622	0.0091	Not normal
	Val_spe	0.127	0.0727	Normal
	Val_dice	0.1746	0.0039	Not normal
VB Light-UNet (proposed method)	Val_sen	0.2353	0.0001	Not normal
	Val_spe	0.2265	0.0001	Not normal
	Val_dice	0.2284	0.0001	Not normal

Bouget et al. [33], which is 0.8, but the MCC obtained by this study is still below 0.9. In addition, the study by Bouget et al. [33] performed segmentation to differentiate brain tumor types, rather than to segment structural features of the tumor, such as the enhancing tumor, nonenhancing tumor, and peritumoral edema. Although the proposed model achieves a Cohen's kappa value of 0.52, this result still reflects a moderate level of agreement, which is considered reasonable for heterogeneous multiclass tumor segmentation tasks. The relatively moderate Kappa score may be attributed to the similarity between certain tumor features and the wide variability in tumor size, which makes accurate segmentation more challenging and can lead to disagreements between predicted and ground truth labels. In addition, the multiclass nature of the task and the presence of class imbalance further contribute to reduced agreement beyond chance, as measured by Kappa. Previous studies have also reported that in multiclass 3D tumor segmentation, even expert annotators typically achieve only moderate agreement due to high inter-rater variability and overlapping morphological characteristics among tumor subregions [27, 28]. Therefore, the Kappa value obtained in this study is consistent with typical reliability levels observed in complex brain tumor segmentation tasks. These findings indicate that, despite the inherent challenges, the proposed model is able to achieve stable and reliable segmentation performance. The high MCC value of 0.96 indicates that the proposed model achieves a strong overall correlation with the ground truth, demonstrating reliable segmentation performance across all classes. The performance evaluation results summarized in Table 8 demonstrate an excellent level of correspondence between the segmented voxels and the ground truth voxels. Based on the overall comparison of brain tumor segmentation performance using the VB Light-UNet architecture with other studies, as presented in Tables 6, 7, and 8, the VB Light-UNet architecture is able to segment brain tumors effectively, and the obtained segmentation results exhibit a high level of similarity with the ground truth. Nevertheless, further development is still required to achieve more optimal performance.

The high performance achieved by the proposed VB Light-UNet was validated through comparisons with several existing studies, as summarized in Tables 6, 7, and 8. Despite the differences in datasets and evaluation metrics used in previous works, the proposed architecture consistently outperformed or achieved comparable results, indicating that the improvements are realistic and not dataset-specific. Furthermore, the superior results can be attributed to the architectural optimizations, including bridge removal, the integration of BN, and the use of pooling indices, which collectively enhance generalization and segmentation accuracy. The proposed model has potential clinical applicability, as it can assist radiologists by providing accurate segmentation of brain tumor subregions in 3D MRI scans, which may support diagnosis, treatment planning, and monitoring of tumor progression. Although the proposed VB Light-UNet architecture demonstrates strong segmentation performance, several limitations remain. The removal of the bridge section, while reducing computational complexity, may limit the model's capacity to capture deeper contextual information. Additionally, the model may be prone to overfitting when trained on limited or unbalanced data, as its performance heavily depends on data

TABLE 5 | Friedman and Wilcoxon test results.

Metric	Friedman test		Wilcoxon test		
	p value	Result	Method	*p* value	Result
Sensitivity	$< 0.001^*$	Significant	V-UNet vs V-UNet without bridge	$< 0.001^*$	Significant
			V-UNet vs VB Light-UNet	$< 0.001^*$	
			V-UNet without bridge vs VB Light-UNet	$< 0.001^*$	
Specificity	$< 0.001^*$	Significant	V-UNet vs V-UNet without bridge	$< 0.001^*$	Significant
			V-UNet vs VB Light-UNet	$< 0.001^*$	
			V-UNet without bridge vs VB Light-UNet	$< 0.001^*$	
Dice score	$< 0.001^*$	Significant	V-UNet vs V-UNet without bridge	0.00585	Significant
			V-UNet vs VB Light-UNet	$< 0.001^*$	
			V-UNet without bridge vs VB Light-UNet	$< 0.001^*$	

*Statistical significance at $p < 0.001$.

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/1114045 by Bangladesh Hinari NPL, Wiley Online Library on [19/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 6 | Comparison of performance evaluation results with other studies on the BraTS 2020 dataset.

Method	Label	Acc	Sen	Spe
FrCN-Net [29]	Tumor cells and background	0.86	0.8	0.89
3D FCN [21]	Enhancing tumor, tumor core, whole tumor, and background	—	0.8	**0.9**
MobileNetV2 [30]	Enhancing tumor, tumor core, whole tumor, and background	—	0.75	0.76
3D-Unet [31]	Enhancing tumor, peritumoral edema, nonenhancing tumor, and background	—	0.83	**0.9**
VB Light-UNet (proposed method)	Enhancing tumor, peritumoral edema, nonenhancing tumor, and background	**0.98**	**0.96**	0.89

Note: The values in bold are the highest.

TABLE 7 | Comparison of Dice score, G-mean, and IoU with other studies on the BraTS 2020 dataset.

Method	Label	Dc	*G*-mean	IoU
FrCN-Net [29]	Tumor cells and background	0.79	0.84	—
3D FCN [21]	Enhancing tumor, tumor core, whole tumor, and background	0.8	0.84	—
MobileNetV2 [30]	Enhancing tumor, tumor core, whole tumor, and background	0.65	0.75	0.62
3D-Unet [31]	Enhancing tumor, peritumoral edema, nonenhancing tumor, and background	0.83	0.86	—
VB Light-UNet (proposed method)	Enhancing tumor, peritumoral edema, nonenhancing tumor, and background	**0.84**	**0.93**	**0.74**

Note: The values in bold are the highest.

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/1114045 by Bangladesh Hinari NPL, Wiley Online Library on [19/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 8 | Comparison of Cohen's kappa and MCC with other studies on the BraTS 2020 dataset.

Method	Label	K	MCC
FrCN-Net [29]	Tumor cells and background	—	0.69
Saliency KSRBN-Net [32]	Tumor cells and background	0.74	0.68
AGU-Net [33]	Glioblastoma, LGG, meningioma, and metastasis	**0.8**	0.85
VB Light-UNet (proposed method)	Enhancing tumor, peritumoral edema, nonenhancing tumor, and background	0.52	**0.96**

Note: The values in bold are the highest.

diversity. Some discrepancies may also occur in regions with weak contrast or irregular tumor boundaries, which can lead to minor oversegmentation or undersegmentation. Therefore, future improvements may involve incorporating attention mechanisms, hybrid CNN–transformer blocks, or advanced regularization strategies to further enhance robustness and generalization across diverse MRI datasets. In addition, data complexity and privacy are important considerations in medical imaging applications. The complexity of brain MRI data, characterized by high variability and heterogeneous tumor structures, can influence model performance and generalization [35]. Furthermore, data privacy remains a critical concern when handling sensitive medical data, and privacy-preserving approaches have been explored in recent studies.

4 | Conclusion

Based on the results and discussion, it can be concluded that the proposed VB Light-UNet architecture is capable of segmenting brain tumor regions in MRI images with high precision and accuracy, as demonstrated by excellent performance across various metrics, including accuracy, IoU, sensitivity, specificity, Dice score, G-mean, and MCC. Despite the relatively low Cohen's Kappa value, the overall results indicate that the model is robust and effective for brain tumor segmentation tasks. The 3D visualization results further confirm that the predicted segmentation closely approximates the ground truth, although some fine-grained details are not fully captured. Although the proposed VB Light-UNet shows promising performance, certain limitations remain. The architectural simplifications introduced to reduce computational complexity may limit the model's ability to capture deeper contextual information, and performance may be affected when applied to limited or highly variable datasets. Future work will focus on incorporating attention mechanisms, hybrid CNN–transformer architectures, and enhanced regularization strategies to improve robustness and generalization. In addition, further studies may explore the integration of the proposed model with other CNN-based architectures and extend its application to classification tasks based on the obtained segmentation results, enabling broader applicability in real-world medical scenarios. Overall, the VB Light-UNet architecture shows strong potential for clinical applications in assisting radiologists with diagnosis, treatment planning, and monitoring of tumor progression.

Acknowledgments

The authors thank the deep learning discussion group and Computation Laboratory, Mathematics, and Natural Science Faculty, Universitas Sriwijaya.

Funding

No funding was received for this manuscript.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets used in this study are publicly available from the Multimodal Brain Tumor Segmentation (BraTS) 2019 and 2020 Challenges, organized by the Medical Image Computing and Computer-Assisted Intervention (MICCAI). The datasets can be accessed through the BraTS Challenge data repository subject to the applicable data access requirements.

References

1. H. A. Khan, W. Jue, M. Mushtaq, and M. U. Mushtaq, "Brain Tumor Classification in MRI Image Using Convolutional Neural Network," *Mathematical Biosciences and Engineering* 17, no. 5 (2020): 6203–6216.

2. L. D, P. C, M. Batumalay, and K. M R, "Comparative Study of CNN-Based Architectures for Early Brain Tumor Diagnosis," *Journal of Applied Data Sciences* 7, no. 1 (2026): 529–540, https://doi.org/10.47738/jads.v7i1.920.

3. I. G. S. M. Diyasa, D. S. Alhamda, H. Sukri, et al., "Optimizing the Classification of Brain Tumors Using Ensemble Learning and Gradient-Weighted Class Activation Mapping on Multi-Section Magnetic Resonance Imaging Images," *International Journal of Technology* 17, no. 2 (2026): 674–691, https://doi.org/10.14716/ijtech.v17i2.8360.

4. P. Wang and A. C. S. Chung, "Relax and Focus on Brain Tumor Segmentation," *Medical Image Analysis* 75 (2022): 102259, https://doi.org/10.1016/j.media.2021.102259.

5. A. Desiani, M. Erwin, B. Suprihatin, S. Yahdin, A. I. Putri, and F. R. Husein, "Bi-Path Architecture of CNN Segmentation and Classification Method for Cervical Cancer Disorders Based on Pap-Smear Images," *International Journal of Computer Science* 48, no. 3 (2021): 1–9.

6. X. Zhao, Y. Wu, G. Song, Z. Li, Y. Zhang, and Y. Fan, "A Deep Learning Model Integrating FCNNs and CRFs for Brain Tumor Segmentation," *Medical Image Analysis* 43 (2018): 98–111, https://doi.org/10.1016/j.media.2017.10.002.

7. T. Nguyen-Phuoc, C. Li, Y. L. Yang, and S. Balaban, "Rendernet: A Deep Convolutional Network for Differentiable Rendering from 3D Shapes," *Advances in Neural Information Processing Systems* (2018): 7891–7901.

8. S. Feng, C. Zuo, W. Yin, G. Gu, and Q. Chen, "Micro Deep Learning Profilometry for High-Speed 3D Surface Imaging," *Optics and Lasers in Engineering* 121, no. April (2019): 416–427, https://doi.org/10.1016/j.optlaseng.2019.04.020.

9. A. Desiani, S. Priyanta, I. Ramayanti, et al., "Multi-Stage CNN: U-Net and Xcep-Dense of Glaucoma Detection in Retinal Images," *Journal of Electronics, Electromedical Engineering, and Medical Informatics* 5, no. 4 (2023): 211–222, https://doi.org/10.35882/jeeemi.v5i4.314.

10. T. M. M. Aung and A. A. Khan, "Enhanced U-Net With Attention Mechanisms Representation in Lung Nodule Segmentation," *Current Medical Imaging* 21 (2025): 1–24.

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/1114045 by Bangladesh Hinari NPL, Wiley Online Library on [19/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

11. D. Rastogi, P. Johri, M. Donelli, et al., "Deep Learning-Integrated MRI Brain Tumor Analysis: Feature Extraction, Segmentation, and Survival Prediction Using Replicator and Volumetric Networks" (2025).

12. D. Rastogi, P. Johri, S. Kadry, et al., *XcepFusion for Brain Tumor Detection Using A Hybrid Transfer Learning Framework with Layer Pruning and Freezing* (2026), 1–30.

13. R. Raza, U. I. Bajwa, Y. Mehmood, M. W. Anwar, and M. H. Jamal, "dResU-Net: 3D Deep Residual U-Net Based Brain Tumor Segmentation from Multimodal MRI," *Biomedical Signal Processing and Control* 79, no. P1 (2023): 103861.

14. T. Lei, W. Zhou, Y. Zhang, R. Wang, H. Meng, and A. K. Nandi, "Lightweight V-Net for Liver Segmentation," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (IEEE, 2020), 1379–1383.

15. O. Cicek, A. Abdulkadir, S. S. Lienkamp, T. Brox, and O. Ronneberger, "3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation," in *Medical Image Computing and Computer-Assisted Intervention* (2016), 424–432.

16. R. Sherman, "A Volumetric Convolutional Neural Network for Brain Tumor Segmentation," in *Comput. Vis. Pattern Recognit* (2018), 1–19.

17. R. Hua, Q. Huo, Y. Gao, et al., "Segmenting Brain Tumor Using Cascaded V-Nets in Multimodal MR Images," *Frontiers in Computational Neuroscience* 14 (2020): 9, https://doi.org/10.3389/fncom.2020.00009.

18. A. Desiani, Erwin, B. Suprihatin, F. Efriliyanti, M. Arhami, and E. Setyaningsih, "VG-DropDNet A Robust Architecture for Blood Vessels Segmentation on Retinal Image," *IEEE Access* 10, no. August (2022): 1–17, https://doi.org/10.1109/access.2022.3202890.

19. R. Rudiansyah, A. Desiani, D. P. Rini, and L. I. Kesuma, "DDUSeg-Net as a Design of Convolutional Neural Network Architecture for Semantic Segmentation in Cervical Cancer," *Int. J. Intell. Eng. Syst.* 18, no. 1 (2025): 980–991.

20. A. Desiani, E. Erwin, B. Suprihatin, Y. Wahyudi, E. S. Cahyono, and M. Arhami, "BVU-Net: A U-Net Modification by VGG-Batch Normalization for Retinal Blood Vessel Segmentation," *International Journal of Intelligent Engineering and Systems* 15, no. 6 (2022): 303–314.

21. J. Sun, Y. Peng, Y. Guo, and D. Li, "Segmentation of the Multimodal Brain Tumor Image Used the Multi-Pathway Architecture Method Based on 3D FCN," *Neurocomputing* 423 (2021): 34–45, https://doi.org/10.1016/j.neucom.2020.10.031.

22. M. El Adoui, S. A. Mahmoudi, M. A. Larhmam, and M. Benjelloun, "MRI Breast Tumor Segmentation Using Different Encoder and Decoder CNN Architectures," *Computers* 8, no. 3 (2019): 1–11, https://doi.org/10.3390/computers8030052.

23. V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNe: a Deep Convolutional Encoder-Decoder Architecture for Image Segmentation," *IEEE Transactions on Pattern Analysis and Machine Intelligence* 39, no. 12 (2017): 2481–2495, https://doi.org/10.1109/tpami.2016.2644615.

24. A. Bahrami, A. Karimian, E. Fatemizadeh, H. Arabi, and H. Zaidi, "A New Deep Convolutional Neural Network Design with Efficient Learning Capability: Application to CT Image Synthesis from MRI," *Medical Physics* 47, no. 10 (2020): 5158–5171, https://doi.org/10.1002/mp.14418.

25. S. Bakas, et al., "Data Descriptor: Advancing the Cancer Genome Atlas Glioma MRI Collections with Expert Segmentation Labels and Radiomic Features," *Nat. Publ. Gr.* 4, no. 170117 (2017): 1–14.

26. B. H. Menze, et al., "The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS)," *IEEE Transactions on Medical Imaging* 99, no. 10 (2014): 1–33.

27. F. Yang, G. Zamzmi, S. Angara, et al., "Assessing Inter-Annotator Agreement for Medical Image Segmentation," *IEEE Access* 11 (2023): 21300–21312, https://doi.org/10.1109/access.2023.3249759.

28. B. Shirokikh, A. Dalechina, A. Shevtsov, E. Krivov, V. Kostjuchenko, and A. Durgaryan, "Systematic Clinical Evaluation of A Deep Learning Method for Medical Image Segmentation: Radiosurgery Application," *Image Video Process* (2021): 1–10.

29. V. V. S. Sasank and S. Venkateswarlu, "An Automatic Tumour Growth Prediction Based Segmentation Using Full Resolution Convolutional Network for Brain Tumour," *Biomedical Signal Processing and Control* 71 (2022): 1–11, https://doi.org/10.1016/j.bspc.2021.103090.

30. M. U. Saeed, G. Ali, W. Bin, et al., "RMU-Net: A Novel Residual Mobile U-Net Model for Brain Tumor Segmentation From MR Images," *Electron* 10, no. 16 (2021): 1–17, https://doi.org/10.3390/electronics10161962.

31. T. Henry, A. Carré, M. Lerousseau, et al., "Brain Tumor Segmentation With Self-Ensembled, Deeply-Supervised 3D U-Net Neural Networks: A BraTS 2020 Challenge Solution," in *Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries* (2021), 327–339.

32. A. Nanda, R. C. Barik, and S. Bakshi, "SSO-RBNN Driven Brain Tumor Classification with Saliency-K-Means Segmentation Technique," *Biomedical Signal Processing and Control* 81 (2023): 104356, https://doi.org/10.1016/j.bspc.2022.104356.

33. D. Bouget, A. Pedersen, A. S. Jakola, et al., "Preoperative Brain Tumor Imaging: Models and Software for Segmentation and Standardized Reporting," *Frontiers in Neurology* 13, no. July (2022): 13–18, https://doi.org/10.3389/fneur.2022.932219.

34. D. F. Brianna, L. I. Kesuma, D. Geovani, and P. Sari, "Combination of Image Enhancement and Double U-Net Architecture for Liver Segmentation in CT-Scan Images," *Journal of Electronics, Electromedical Engineering, and Medical Informatics* 7, no. 1 (2025): 208–219.

35. A. Kujur, Z. Raza, A. A. Khan, and C. Wechtaisong, "Data Complexity Based Evaluation of the Model Dependence of Brain MRI Images for Classification of Brain Tumor and Alzheimer's Disease," *IEEE Access* 10, no. September (2022): 112117–112133, https://doi.org/10.1109/access.2022.3216393.

Author Biographies

Anita Desiani was born in Palembang, Indonesia, in 1977. In 2020, she is currently working on a project for her Doctoral Program at the Mathematics and Natural Science Faculty, Universitas Sriwijaya. She received her mathematics bachelor from Universitas Sriwijaya in 2000 and magister degree in Computer Science from Universitas Gadjah Mada in 2003. In the same year, she joined as a lecturer at the Mathematics Department in Universitas Sriwijaya until now. Her current research is in the field of Data Mining, image processing, pattern recognition, computer vision, and artificial intelligence. She is also the author of the books *Artificial Intelligence* and *MATLAB Programming*.

Deshinta Arrova Dewi is an esteemed educator and researcher, renowned for her extensive expertise in Artificial Intelligence, Data Science, and Software Engineering. She received her Ph.D. from the National University of Malaysia (UKM) in 2019. With over 2 decades of dedicated service in tertiary education across Malaysia and Indonesia, Dr. Dewi has cultivated a deep understanding of these complex fields. Her dynamic career is highlighted by her leadership in guiding a prominent research group that collaborates on projects with various industry partners. Her drive and enthusiasm make her a pivotal figure in academic and research circles, consistently pushing the boundaries of technology and innovation.

Dite Geovani was born in Palembang in January 2002. She is currently working on a project for the Doctoral Program at the Engineering Faculty, Universitas Sriwijaya. She received a bachelor's degree in Mathematics from Universitas Sriwijaya in 2023 and is currently undergoing a PMDSU scholarship program at Universitas Sriwijaya since 2023. Her current research includes the fields of image processing, pattern recognition, data mining, and artificial intelligence.

Muhammad Arhami was born in Pulo Leun Teuga, October 1974. He received mathematics bachelor from Universitas Syahkuala in 2000, and magister degree in Computer Science from Universitas Gadjah Mada in

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/1114045 by Bangladesh Hinari NPL, Wiley Online Library on [19/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

2004. He has been working at Politeknik Negeri Lhokseumawe since 2001. He became an Associate Professor in 2013. His research is in artificial intelligence, mathematics, data mining, software engineering, and data structures. He already has experience as an author of artificial intelligence, expert systems, MATLAB programming books, and mathematics books.

Emy Setyaningsih was born in Semarang, Indonesia, in 1972. She received her Bachelor of Computer Science from IST AKPRIND, Indonesia, in 1996, M.Kom. degree in Computer Science from the Universitas Gadjah Mada (UGM), Yogyakarta, Indonesia, in 2004, and a Doctor's degree in Computer Science from the Universitas Gadjah Mada (UGM), Yogyakarta, Indonesia, in 2019. In 1997, she joined AKPRIND University as a Lecturer. She was an Associate Professor in 2009. Her current research interests are cryptography, digital image processing, and pattern recognition.

Lucky Indra Kesuma was born in Palembang on September 25, 1990. He completed his education with an Associate Degree in Computerized Accounting in 2011 at Sriwijaya University. He continued his education for a Bachelor's degree in Information Systems, graduating in 2014 from Sriwijaya University. He then pursued a Master's degree in Computer Science, specializing in Informatics Engineering, completing his studies in 2017 at Bina Darma University. In 2023, he finished his doctoral studies (Ph.D.) in Engineering Science (Informatics). The author is a lecturer at the Department of Electrical Engineering, Universitas Sriwijaya, Inderalaya.

4795, 2026, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/acis/1114045 by Bangladesh Hinari NPL, Wiley Online Library on [19/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License